Risk Documentation

Subgroup 3: Device

Revision History

Date	Version	Description	Author
2026-02-07	1.0	Initial Risks	Patrick Nordahl, Tony Nguyen
2026-02-28	1.1	Added R5 Communication Failure Risk	Patrick Nordahl
2026-03-15	1.2	Added **prevention** strategies for concurrency and race conditions.	Tony Nguyen, Patrick Nordahl
2026-04-13	1.3	Added R6 Audio Playback Blocking System Execution	Sergej Macut
2026-05-08	1.4	Updated risk mitigations following the Node.js gateway refactor, C++ architectural overhaul, and implementation of the Observer Pattern for autonomous logic.	Mustafa Al-Bayati Ryad Hazin Sergej Macut Tony Nguyen

Risk List

Risk Description	Priority
R1. Changing requirements during the project	High
R2. Issues with integration between device simulations and other system components	High
R3. Limited Experience with certain technologies used in the project	Medium
R4. Scope Expansion	High
R5. Communication Failure / Race Conditions	Critical/High
R6. System Blocking during Execution	Medium

Risk Handling Plans

R1. Risk description

As the project progresses, the understanding of device behavior and system interaction might improve, leading to changes in the initial requirements definitions.

Probability: High

Impact: Medium

Handling Plan:
Requirements will be treated as a living document and updated as the project progresses.

R2. Risk Description:

Integration issues might arise between the simulated devices and other system components due to mismatched assumptions or interface definitions

Probability: Medium

Impact: High

Handling Plan:
Early alignment on interfaces and incremental integration testing will be used to reduce integration risks

R3. Risk Description:

The subgroup may have limited prior experience with certain technologies used for the project, which may slow development or cause implementation issues

Probability: Medium

Impact: Medium

Handling Plan:
The group will implement functionality incrementally and validate each step before proceeding, reducing the risk of major implementation issues.

R4. Risk Description:

There is risk that additional features or devices are introduced beyond the initial defined scope, increasing complexity and development effort

Probability: Medium - High

Impact: High

Handling Plan:
Clear prioritization of essential requirements will be maintained, and scope changes will be evaluated and agreed upon by the group before implementation.

R5 .Risk Description

Simultaneous data streams from the Arduino (observations) and the Server (commands) can cause buffer overflows or race conditions.

Handling Plan:

- **Node.js Migration:** Ported the gateway from Python to Node.js to utilize its non blocking, event driven I/O, ensuring that serial messages do not block user command processing.
- **Status Feedback Loop:** The device transmits an `ACK` signal after every state change to keep the UI synchronized with the actual hardware state.

R6. System Blocking during Execution

- **Risk Description:** Heavy processes, specifically audio playback, can block the CPU and prevent the house from reacting to hazards like gas or steam.
- **Handling Plan:**
    - **Encapsulated MusicEngine:** Refactored audio logic into a dedicated class that uses non-blocking timing logic, allowing the `music.update()` function to run concurrently with sensor checks.
    - **Safety Interruption:** High priority autonomous alerts (via the Observer Pattern) are designed to override or interrupt secondary processes like music playback.